

SUPPLEMENTARY MATERIAL

A. System Conditions and Data

TABLE S1: Kinetic Parameters Used in the Dynamic Simulation of *M. smegmatis* Growth

Parameter	Description	Units
$\mu_{\max}$	Maximum specific growth rate	[1/h]
k_C	Monod constant for glycerol	[g/L]
k_O	Monod constant for oxygen	[g/L]
k_d	Death/decay rate constant	[1/h]
t_{lag}	Lag phase time constant	[h]
$X_{\max}$	Logistic capacity (provisional term)	[g/L]
pH_{opt}	pH optimum (centre of inhibition bell)	[-]
pH_w	Width of the pH-inhibition bell	[-]
$Y_{X/C}$	Biomass yield on glycerol	[g/g]
Y_{X/O_2}	Biomass yield on O_2	[g/g]
Y_{X/CO_2}	Biomass yield on CO_2	[g/g]
m_C	Maintenance coefficient (Pirt)	[g glycerol/(g X · h)]
$Y_{CO_2}^m$	Maintenance stoichiometric factor $\rightarrow CO_2$	[g CO_2 /g glycerol]
$Y_{O_2}^m$	Maintenance stoichiometric factor $\rightarrow O_2$	[g O_2 /g glycerol]
$\rho_{N,\max}$	Maximum specific N-uptake rate	[g N/(g X · h)]
k_{up}	N-uptake half-saturation constant	[g N/L]
$q_{\min}$	Subsistence N quota (growth floor)	[g N/g X]
$q_{\max}$	Quota ceiling (feedback-inhibition parameter)	[g N/g X]
f_{Nrec}	Fraction of quota recycled on cell death	[-]

TABLE S2: Physico-Chemical Constants Used in the Algebraic pH (Charge Balance) System

Constant	Description	Value
pK_{a1}	Phosphate dissociation ($H_2PO_4^-/HPO_4^{2-}$)	6.86
pK_{a2}	Citrate dissociation, step 1 (H_3Cit/H_2Cit^-)	3.13
pK_{a3}	Citrate dissociation, step 2 ($H_2Cit^-/HCit^{2-}$)	4.76
pK_{a4}	Citrate dissociation, step 3 ($HCit^{2-}/Cit^{3-}$)	6.40
pK_{a5}	Ammonium dissociation (NH_4^+/NH_3)	9.25
pK_{a6}	Glycerol dissociation ($C_3H_8O_3$) — <i>not used in current equations</i>	14.15
pK_{a7}	Carbonic acid dissociation ($CO_2(aq)/HCO_3^-$)	6.35
pK_{a8}	Bicarbonate dissociation (HCO_3^-/CO_3^{2-}) — <i>not used, 2nd carbonate step omitted</i>	10.33
pK_{a9}	Water autoionization	14.00
$[KH_2PO_4]$	Total phosphate concentration	2.18
$[C_6H_8O_7]$	Total citrate concentration	2.00
pH_{alk}	Strong-ion background / reference alkalinity	7.20
M_N	Molar mass of nitrogen [g/mol]	14.007

TABLE S3: Gas-Transfer Constants Used in the Oxygen Mass Balance

Constant	Description	Units	Value
O_2^{sat}	O_2 saturation concentration	[g/L]	7.2673×10^{-3}
k_{La}	O_2 volumetric mass-transfer coefficient	[1/h]	86.26

Initial values of variables in normal and ammonium-deficient simulations where those correspondent to the experimental data on each condition.

B. Díaz model in normal conditions

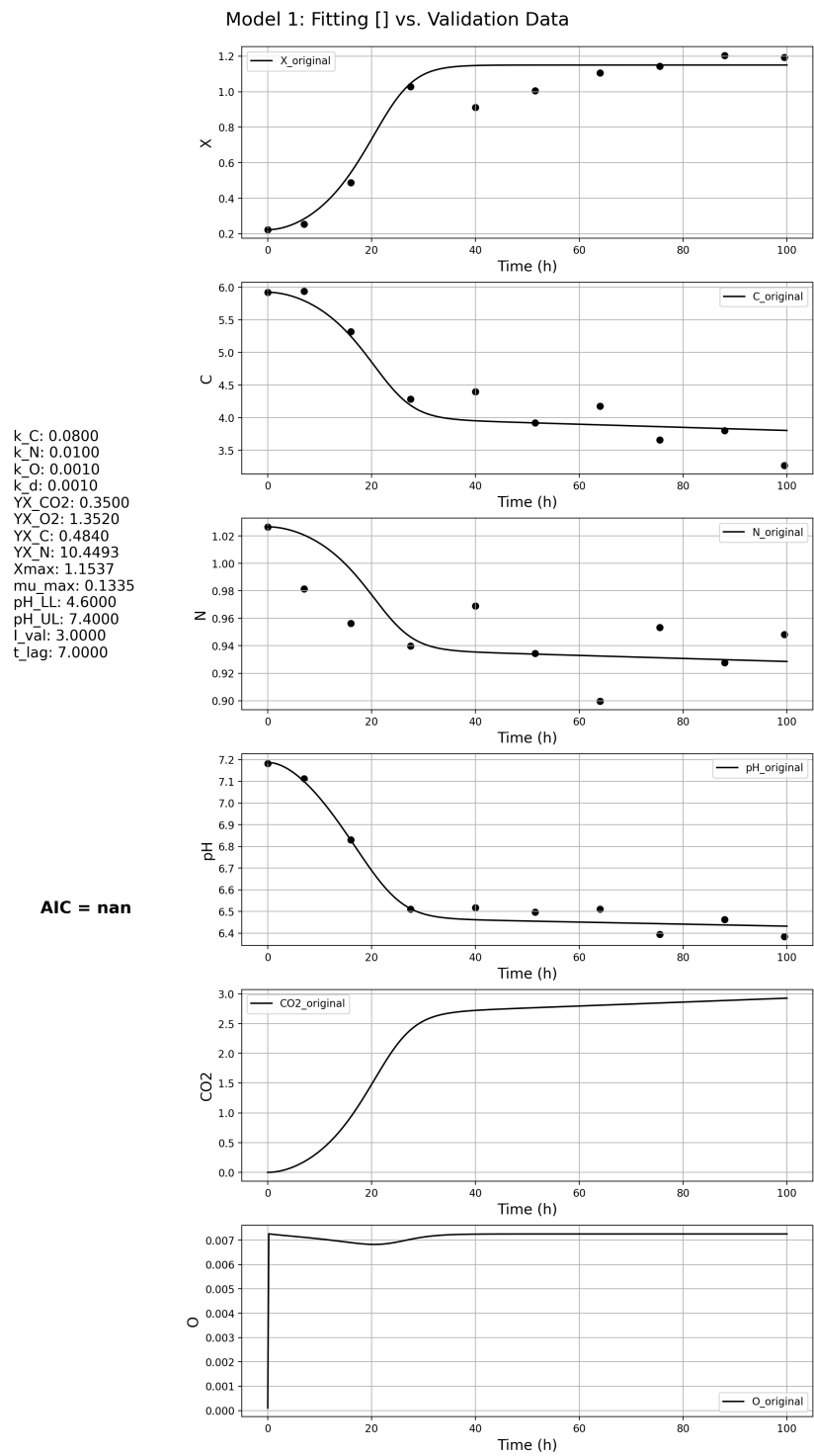


Fig. S1: Variable responses simulated with Díaz model vs experimental data in normal culture conditions

C. Díaz model in ammonium-deficient conditions

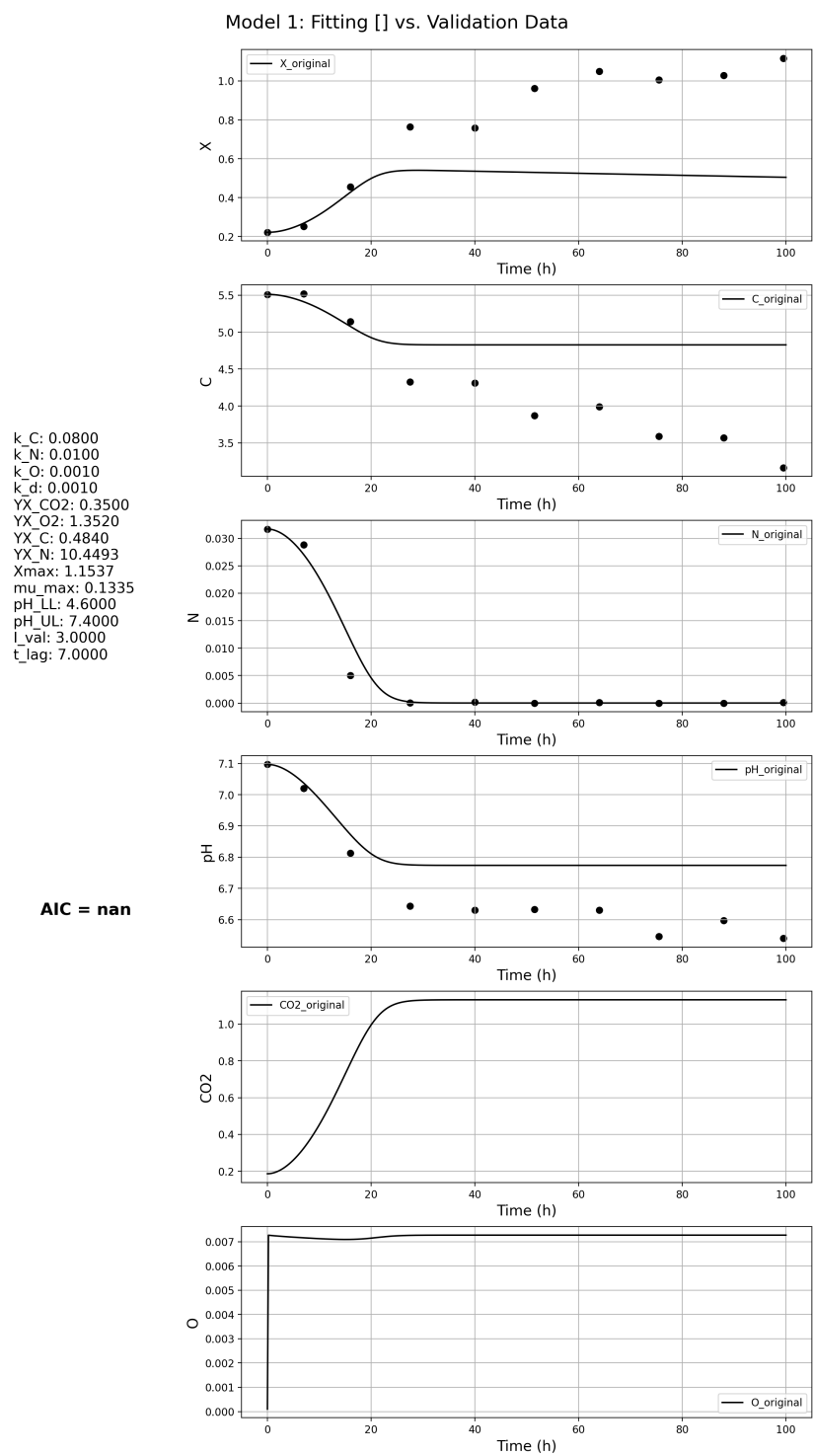


Fig. S2: Variable responses simulated with Díaz model vs experimental data in ammonium-deficient culture conditions

D. Reformulated Droop based model in normal conditions and baseline parameters

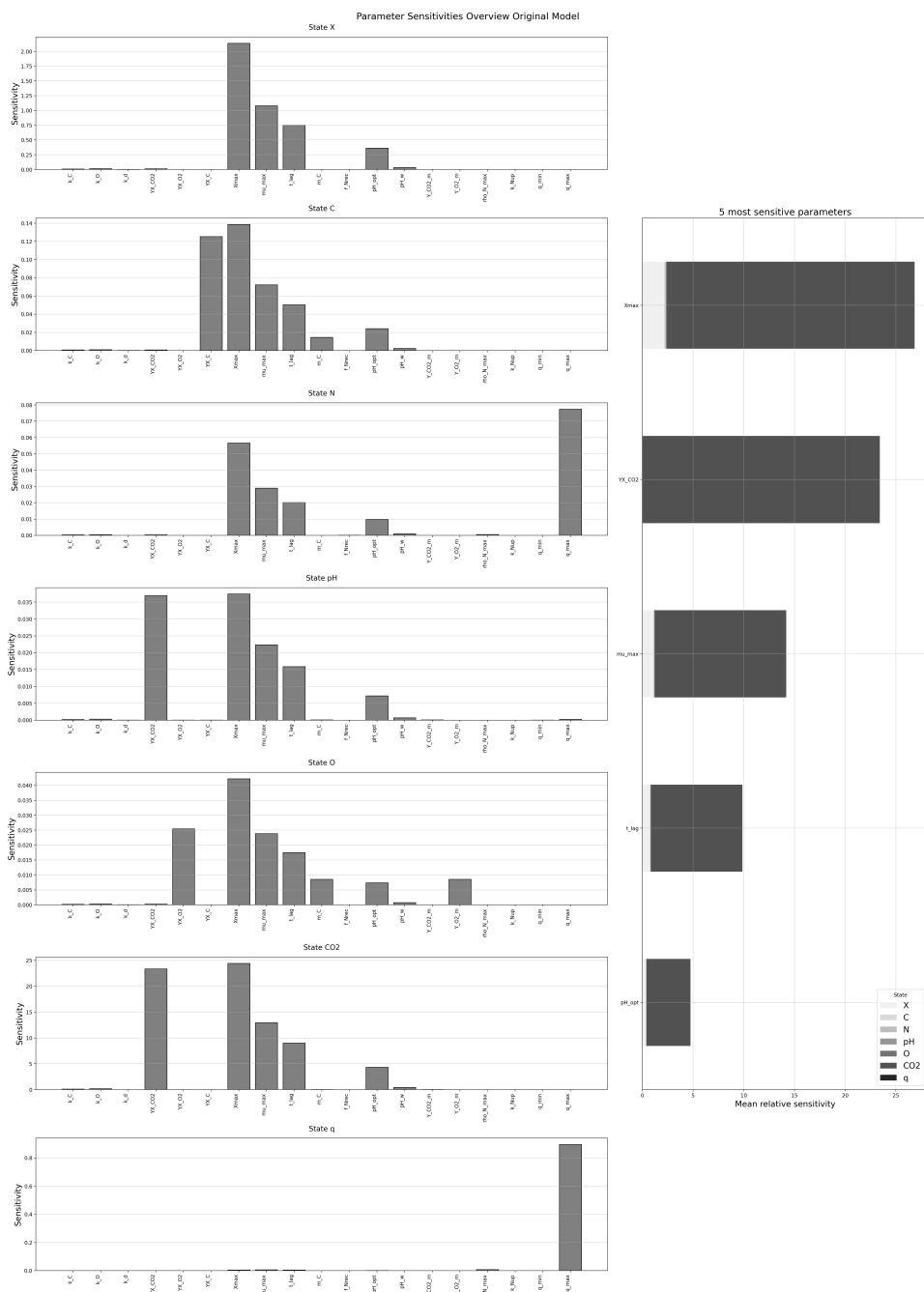


Fig. S3: Parameter sensitivity analysis of the reformulated (Droop-based) model with baseline, uncalibrated parameters, in normal culture conditions

E. Reformulated Droop based model in ammonium-deficient conditions and baseline parameters

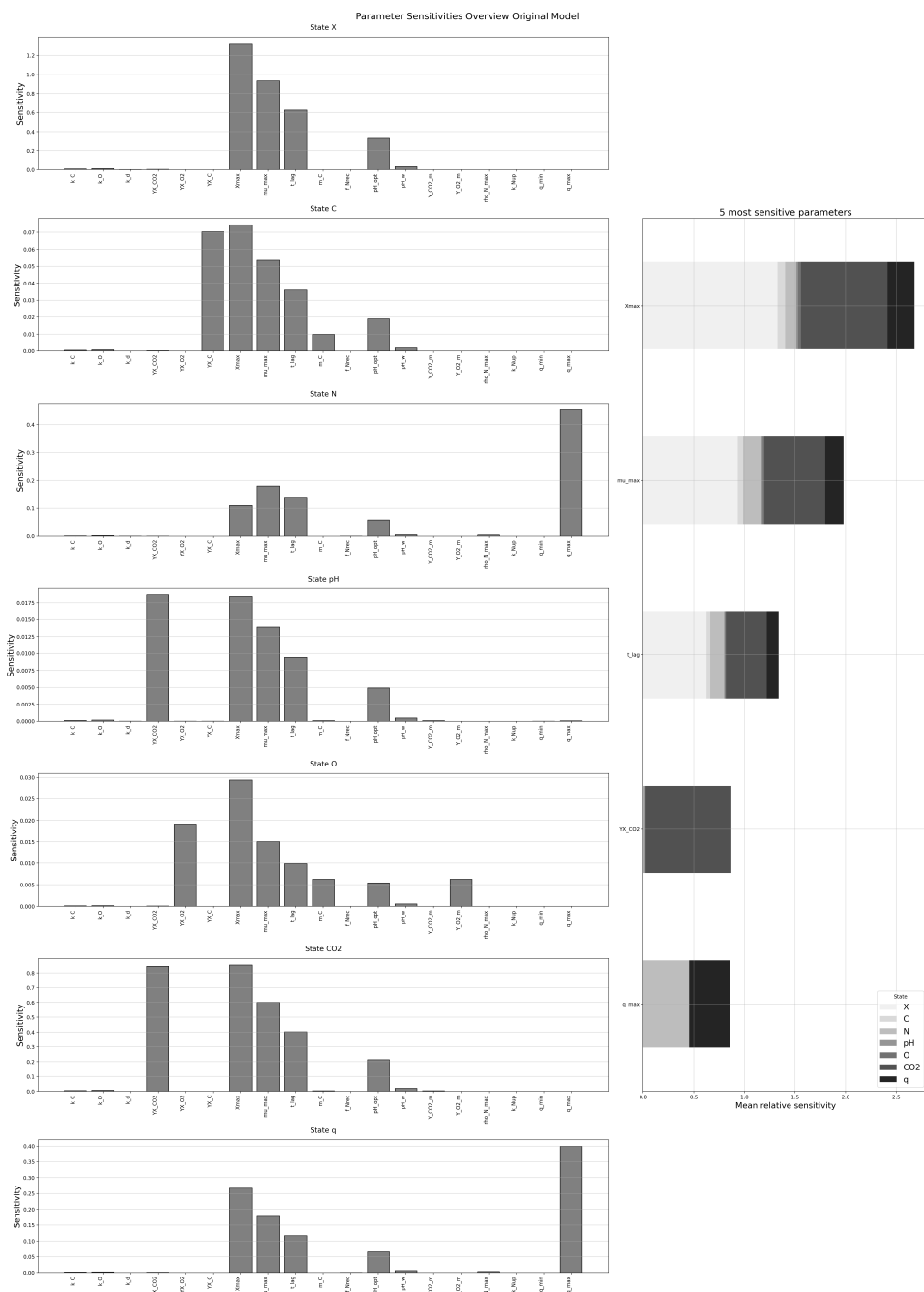


Fig. S4: Parameter sensitivity analysis of the reformulated (Droop-based) model with baseline, uncalibrated parameters, in ammonium-deficient culture conditions

F. Calibrated Droop based model in normal conditions

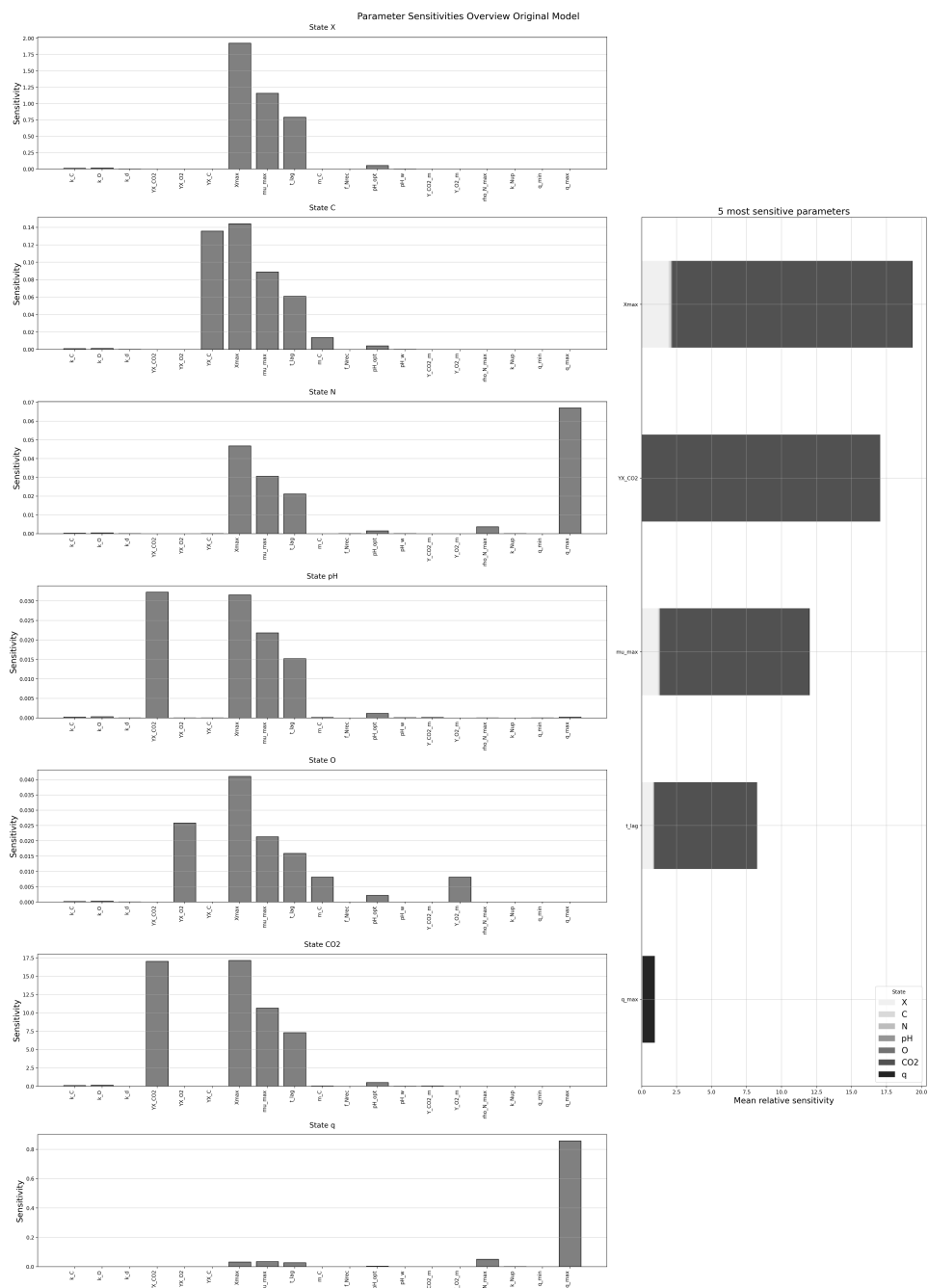


Fig. S5: Parameter sensitivity analysis of the calibrated reformulated model in normal culture conditions

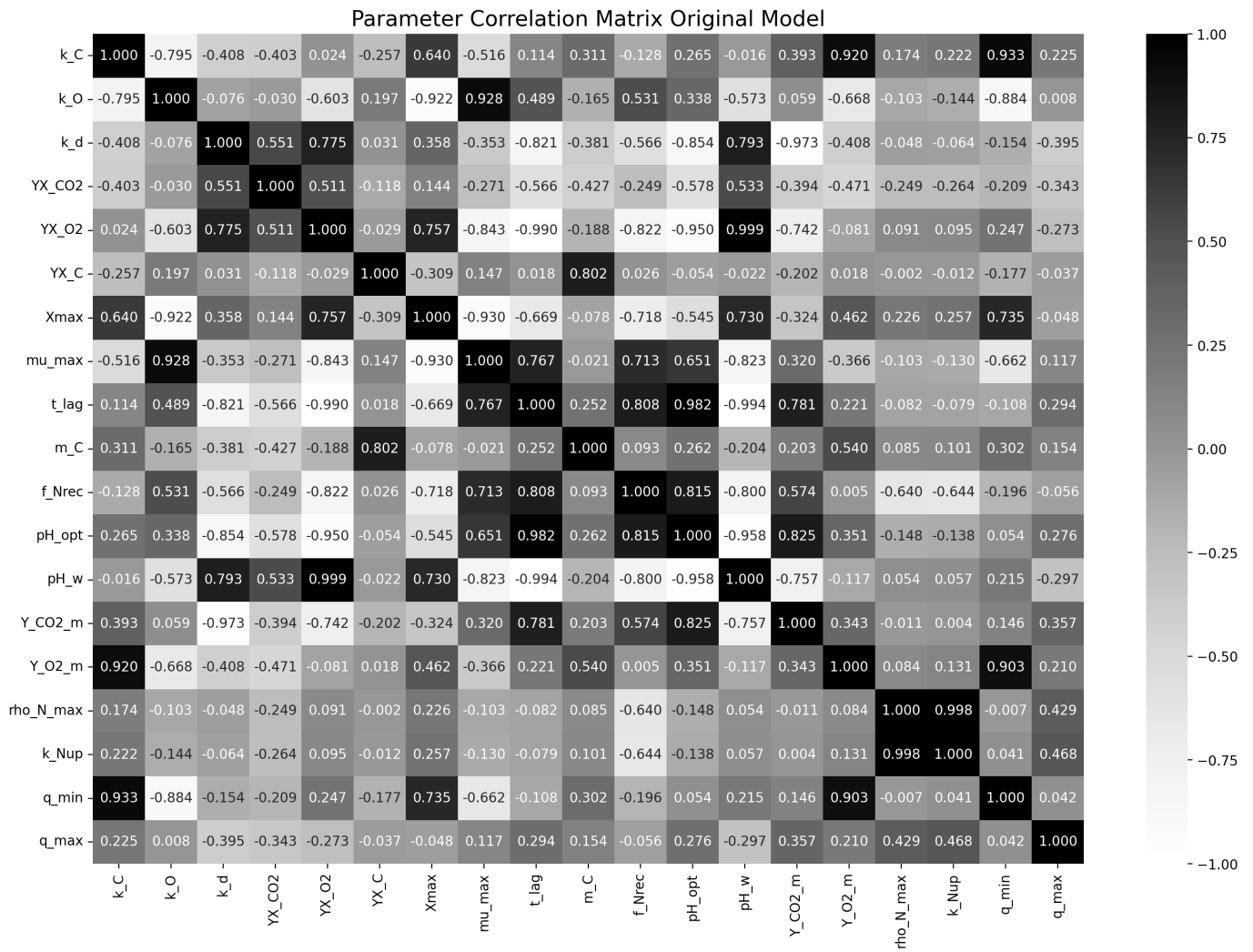


Fig. S6: Pairwise parameter correlation matrix for the calibrated reformulated model in normal culture conditions

Residuals Analysis Original Model

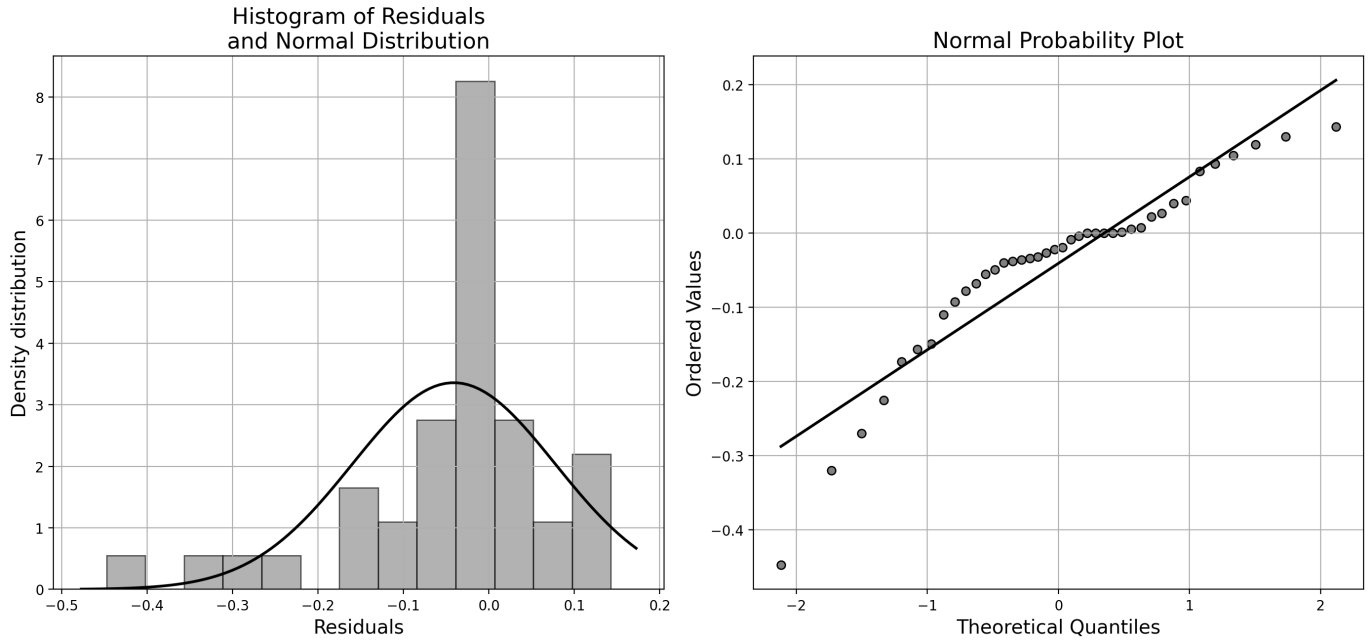


Fig. S7: Residual diagnostics (histogram with fitted normal density, and normal probability plot) for the calibrated reformulated model in normal culture conditions

TABLE S4: Staged calibration record for the reformulated model – normal culture conditions.

Model	Parameters estimated	NRMSE					AIC	BIC
		X	C	N	pH	Global		
Uncalibrated	—	0.0883	0.1126	0.2030	0.0611	0.0228	—	—
Iteration 1	$k_C, k_O, X_{max}, t_{lag}$	0.1460	0.1127	0.2286	0.1424	0.0254	-128.69	14.79
Iteration 2	$k_O, X_{max}, \mu_{max}, t_{lag}$	0.0847	0.0949	0.1974	0.0642	0.0196	-149.46	14.77
Iteration 3	$k_C, k_O, X_{max}, \mu_{max}, t_{lag}$	0.0846	0.0966	0.1978	0.0634	0.0199	-146.21	18.46
Iteration 4*	$k_C, k_O, Y_{X/CO_2}, Y_{X/C}, X_{max}, \mu_{max}, t_{lag}, pH_{opt}$	0.0844	0.0812	0.1970	0.1066	0.0178	-148.85	29.53

* Final calibration stage; corresponds to the Reformulated model, Normal column of Table II in the main text.

G. Calibrated Droop based model in ammonium-deficient conditions

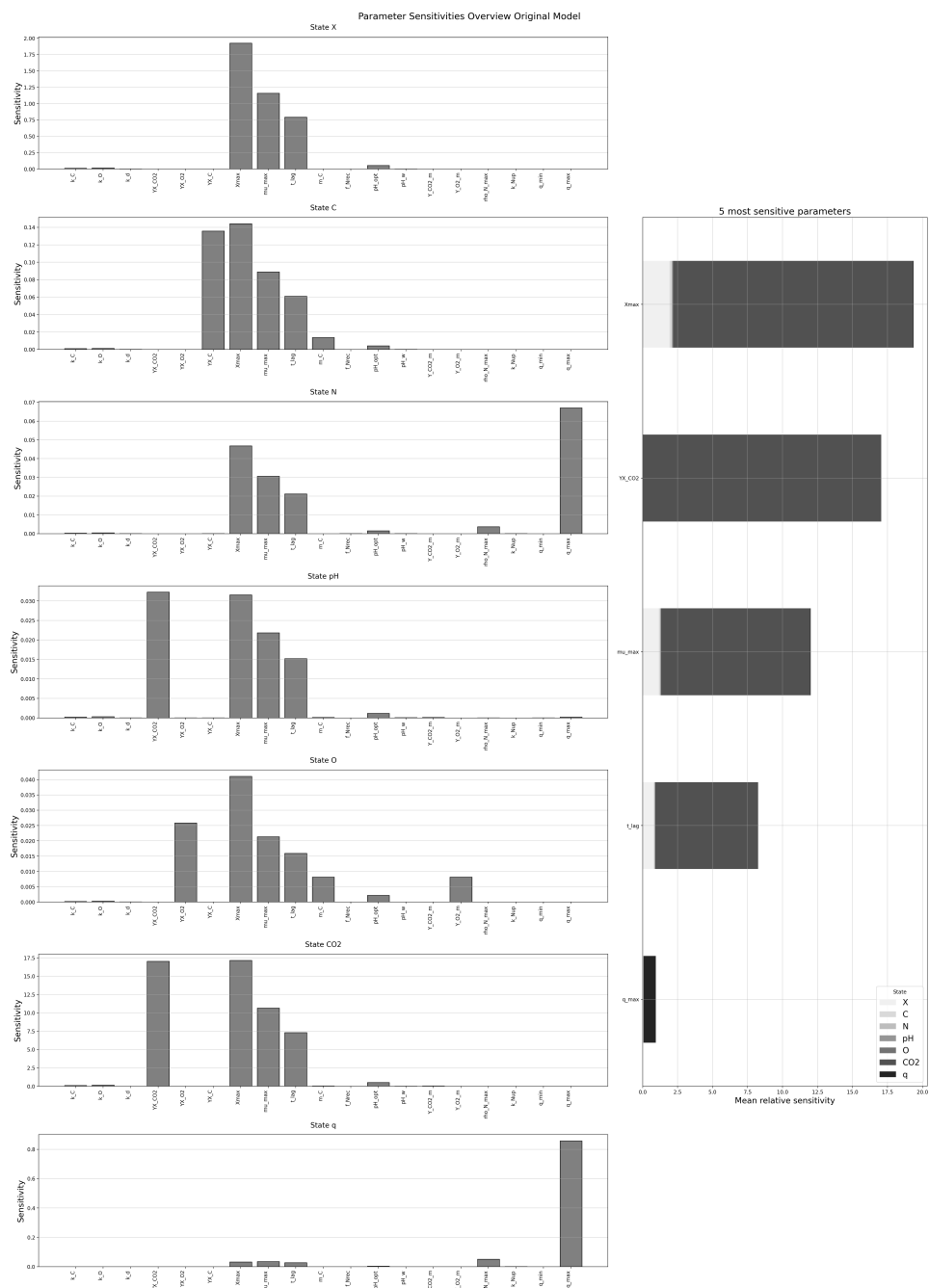


Fig. S8: Parameter sensitivity analysis of the calibrated reformulated model in ammonium-deficient culture conditions

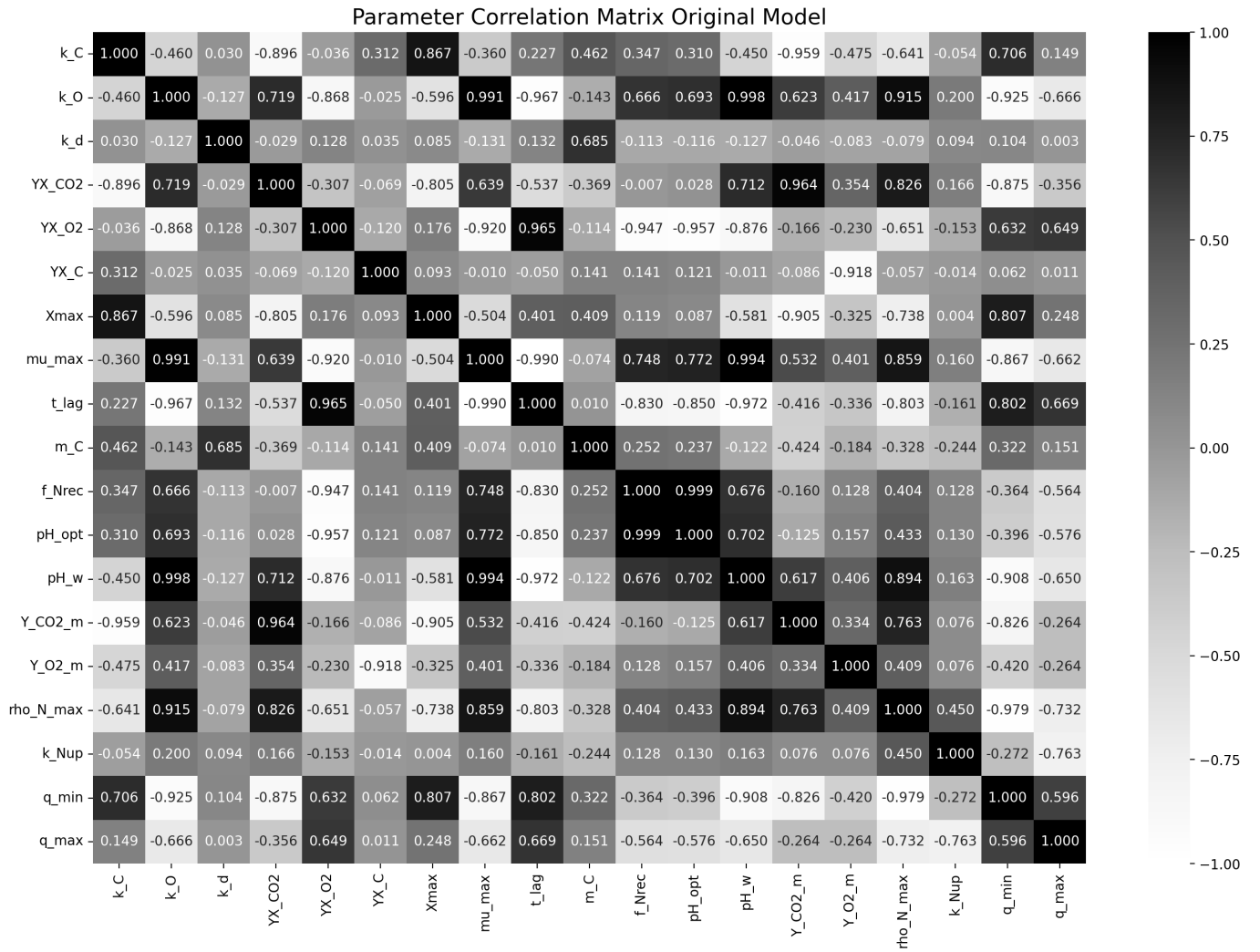


Fig. S9: Pairwise parameter correlation matrix for the calibrated reformulated model in ammonium-deficient culture conditions

Residuals Analysis Original Model

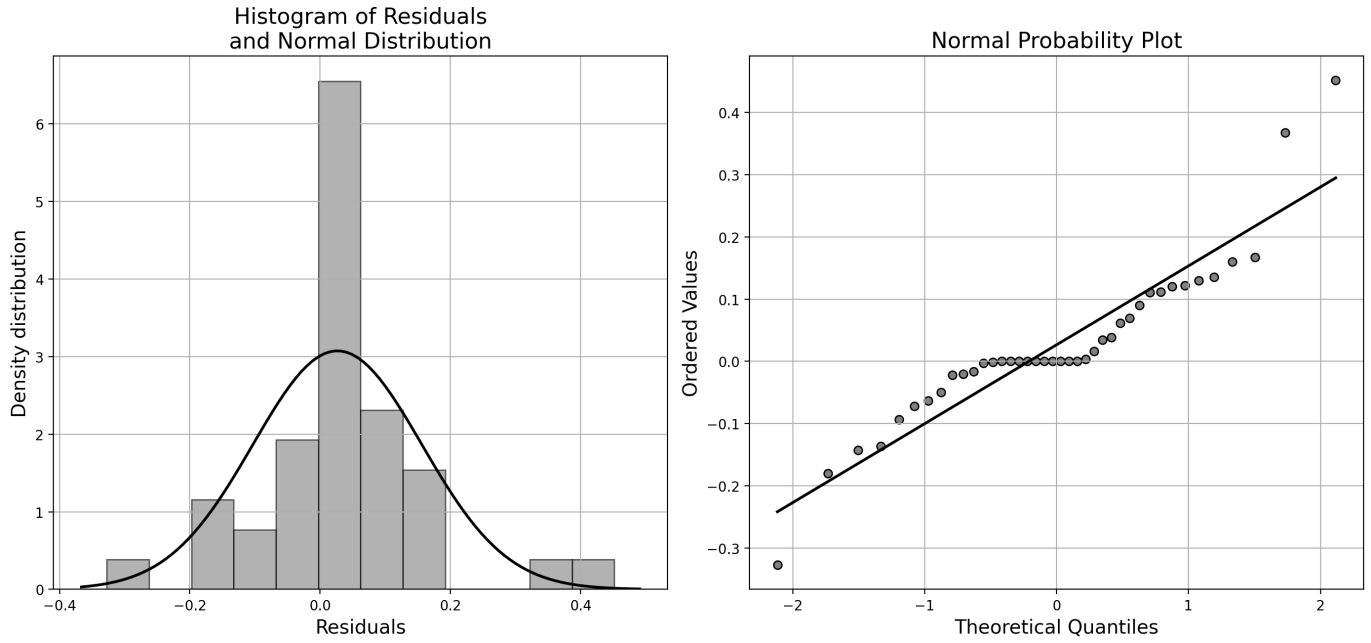


Fig. S10: Residual diagnostics (histogram with fitted normal density, and normal probability plot) for the calibrated reformulated model in ammonium-deficient culture conditions

TABLE S5: Staged calibration record for the reformulated model – ammonium-deficient culture conditions.

Model	Parameters estimated	NRMSE					AIC	BIC
		X	C	N	pH	Global		
Uncalibrated	—	0.1406	0.1987	0.0761	0.2114	0.0352	—	—
Iteration 1	$k_C, k_O, X_{max}, t_{lag}$	0.0990	0.1401	0.0957	0.1932	0.0252	-127.56	14.79
Iteration 2	$k_C, k_O, X_{max}, \mu_{max}, t_{lag}$	0.1488	0.1963	0.0760	0.2043	0.0348	-99.77	18.51
Iteration 3*	$k_C, k_O, Y_{X/CO_2}, Y_{X/C}, X_{max}, \mu_{max}, t_{lag}, pH_{opt}$	0.1524	0.0900	0.0761	0.1279	0.0185	-144.60	29.53

* Final calibration stage; corresponds to the Reformulated model, Ammonium deficiency column of Table II in the main text.